AHMED ALVEE

Solutions Architect | AI & Automation Engineering
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alvee1994.github.io/portfolio

Summary

Solutions architect who built AWS data pipelines and LLM workflows for clinics, and took them live with customers on site. Led the first overseas deployment at Mesh Bio and a team of 2-3 solutions engineers. AWS Certified Solutions Architect – Associate. Completing an MBA at Rotterdam School of Management.

Skills

AWS (Lambda, Step Functions, SQS, SNS, EventBridge), MLLP, Postman, LangGraph, Amazon Bedrock, Python, REST APIs, HL7, DICOM, Terraform, LangChain, Claude Code, Microsoft Graph API, Gemini, Pydantic, ChromaDB, MongoDB, n8n, Cursor, C++, Qt, Linux, Git, ROS, English (fluent)

Professional Experience

Mesh Bio Pte Ltd — Singapore, SG
Solutions Architect (prev. Senior Solutions Engineer) — 05/2022 – 12/2025
Singapore health-tech company serving private healthcare providers. Its health screening workflow scored patient risk and built screening reports. Built the pipelines behind it.

- Cut per-submission time from about 1 hour to 6-10 minutes for 30+ clinics with an AI clean-up pipeline (LangGraph, Claude on Bedrock, Pydantic, ChromaDB); a reviewer checked and submitted each file.
- Cut lab and radiology batch turnaround from about a week to a day by replacing a hardcoded Lambda per clinic and lab with a config-driven Step Functions pipeline handling about 10,000 records an hour in Singapore.
- Kept wrong data out of patient profiles with record count checks and exception handling, and resolved a lab vendor's unannounced system switch in one day.
- Cut onboarding per clinic from 3-5 days to under 1 day by redesigning client onboarding on AWS (Lambda, SNS, SQS, Step Functions, HL7).
- Turned the first overseas customer into a lighthouse account that opened follow-on regional projects by recommending a headless API-based workflow and leading 3 weeks of on-site deployment.
- Owned the integration APIs that became the distribution partner's primary way to integrate new clients; joined technical sales calls to map customer tech stacks.
- Promoted to Solutions Architect; hired, trained and supervised a team of 2-3 solutions engineers who expanded the APIs and pipelines.

Focus areas: AWS Lambda, Step Functions, SQS/SNS, EventBridge, HL7, MLLP, LangGraph, Amazon Bedrock, Pydantic, ChromaDB, team leadership

Creative Medtech Solutions Pte Ltd — Singapore, SG
Software Engineer — 09/2021 – 04/2022
Singapore medtech company building a robotic system that ablates cancer cells with high intensity focused ultrasound (HIFU).

- Owned the annotation and user-interaction layer of the imaging system for a HIFU cancer-ablation robot: marking and measuring ablation targets, and DICOM export.

Focus areas: C++, medical imaging, DICOM, operator GUI, annotation tooling

Singapore University of Technology and Design — Singapore, SG
Research Assistant — 02/2018 – 08/2021

University research lab building robotics and wearable systems, working with teams of research assistants and PhD students.

- Mapped 2 floors and 5 rooms in under 6 minutes with a 12+ robot system, owning its locomotion code.
- Recognized 7 hand gestures to command unmanned vehicles with an ML pipeline (Python, ROS, StretchSense).
- Contributed C++ and hardware engineering to 2 more robotics builds: a legged mapping robot and a custom PCB for 3D-knitted wearables.

Focus areas: C++, Python, ROS, machine learning, multi-robot systems, PCB design, embedded firmware

Consulting Experience

RSM Living Management Project — Rotterdam, Netherlands
AI Strategy Consultant (Client: Truvalu) — 06/2026 – 07/2026

MBA consulting project for Truvalu, a Netherlands-based impact investor supporting agri-food SMEs in emerging markets.

- Projected about a 45% cut in monitoring time for an impact investor with a 5-stage AI monitoring workflow, designed as part of a 5-person MBA team.
- Engineered and demoed AI tools and agents (Gemini, Claude, Tally.so) spanning all 5 workflow stages, from data clean-up to reporting.
- Co-led the final client presentation to Truvalu leadership.

Focus areas: AI workflow design, Gemini, Claude, Tally.so, AI agents, client presentation

Education

Rotterdam School of Management, Erasmus University — Rotterdam, Netherlands
Master of Business Administration (MBA) — 01/2026 – 03/2027

Singapore University of Technology and Design — Singapore, SG
BEng (Honours), Engineering Product Development — 05/2014 – 08/2017

Certifications

- AWS Certified Solutions Architect – Associate, Amazon Web Services (10/2023).